

Experiment no-13

13. Generate a parse tree for a given sentence using a context-free grammar using python program.

PROGRAM:

```
import nltk
from nltk import CFG
from nltk.parse import ChartParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> Pronoun
VP -> V NP
Pronoun -> 'i' | 'you'
V -> 'like'
""")

parser = ChartParser(grammar)

sentence = input("Enter a sentence: ").lower().split()

for tree in parser.parse(sentence):
    print(tree)
```

OUTPUT:

```
... Enter a sentence: i like you
(S (NP (Pronoun i)) (VP (V like) (NP (Pronoun you))))
```